

Graph Attention Design Choices Matter: A Controlled Study of LoRA-Adapted Audio Anti-Spoofing

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Abstract—Audio anti-spoofing systems increasingly combine self-supervised learning, parameter-efficient fine-tuning, and graph-attention-based backends. However, performance gains in such systems are often entangled with concurrent changes in the backbone, fine-tuning strategy, and training protocol, making the independent contribution of graph attention design difficult to isolate. To address this issue, we conduct a systematic controlled study of the graph attention layer under a unified experimental setting. We decompose the layer into three independently testable design dimensions: scoring symmetry, temperature learnability, and routing granularity. These are instantiated as a concat-based scoring branch, a LearnT branch with learnable temperature, and a multi-temperature routing branch, respectively. Each dimension is implemented as an independently gated residual branch, enabling the evaluation of both individual variants and their combinations under the same experimental setting. Experiments on five evaluation sets with five random seeds show that the LearnT branch achieves the best average equal error rate (EER), yielding a 16.1% relative improvement over the baseline. In contrast, the multi-temperature routing branch does not improve average performance on its own, but substantially reduces cross-seed standard deviation when combined with the concat-based scoring branch. Moreover, two individually effective branches degrade performance when used together, resulting in a 25.6% relative deterioration compared with the baseline. This finding reveals strong non-additive interactions among graph attention design dimensions. Overall, the results suggest that, under parameter-constrained fine-tuning, improvements in graph attention layers depend more on capacity allocation and branch interaction than on simply adding more learnable parameters.

Index Terms—anti-spoofing, graph attention, parameter-efficient fine-tuning, attention temperature, capacity allocation

I. INTRODUCTION

Recent advances in generative audio have made synthetic speech increasingly difficult to distinguish from genuine recordings [1], thereby posing serious threats such as fraud and the dissemination of disinformation. In audio anti-spoofing, a

common paradigm combines a self-supervised speech front-end [2]–[5], such as XLS-R [6], with a graph-attention-based backend, such as AASIST2 [7]. Recent parameter-efficient systems further adapt SSL front-ends with LoRA-style modules [8]–[11], while AASIST variants continue to modify the graph-attention backend [12], [13]. In our controlled setting, LoRA modules are inserted into the Transformer layers of the front-end and trained together with LayerNorm parameters, while the original pretrained backbone weights remain frozen, reducing the number of trainable parameters to below 1% of the full model, as illustrated in Figure 1. Although these systems have further extended the overall SSL-plus-backend framework, the graph-attention backend has largely been treated as a fixed component, and its internal design choices have rarely been examined in a direct and systematic manner.

To characterize this gap more precisely, we revisit the graph attention layer in AASIST2. This layer involves three implicit design choices: (i) product-based symmetric scoring (i.e., $e_{ij} = e_{ji}$), (ii) a fixed softmax temperature, and (iii) a single global temperature shared across all nodes. Related work has studied adjacent mechanisms in separate contexts: GATv2 [14] modifies graph-attention scoring, mixture-of-experts methods introduce routing mechanisms for graph neural networks [15], GATS [16] learns node-wise temperatures for GNN calibration, and AASIST3 [13] evaluates fixed-temperature variants in anti-spoofing. However, these factors have not been isolated as controlled design dimensions within LoRA-adapted AASIST2 audio anti-spoofing.

To address this gap, we conduct a controlled factorization of the graph-attention design space under a unified experimental setting. Specifically, we decompose it into three independently testable dimensions: scoring symmetry, temperature learnability, and routing granularity, and instantiate each of them as a corresponding residual branch for systematic comparison. All variants are evaluated under the same adaptation protocol,

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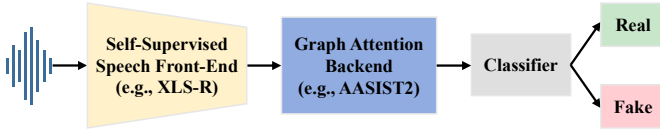


Fig. 1. Overview of the LoRA-adapted audio anti-spoofing framework used in this study.

enabling a more direct analysis of the effects of individual modifications and their combinations.

Systematic evaluations on one in-domain test set, four out-of-domain test sets, and five random seeds reveal pronounced non-additive interactions among these design dimensions. The LearnT branch achieves the best average EER. By contrast, the multi-temperature routing branch does not improve average performance when used in isolation, but yields the lowest cross-seed standard deviation when combined with the concat-based scoring branch. More notably, two modifications that are each effective when introduced separately lead to a substantial performance degradation when applied jointly. Overall, these findings indicate that the effects of different design choices within graph attention are not independent, and that a systematic analysis of their interactions is important for improving current audio anti-spoofing models.

We summarize our contributions as follows:

- 1) We present a controlled decomposition of graph attention design choices in LoRA-adapted audio anti-spoofing into independently testable dimensions.
- 2) We provide empirical evidence that, within a unified LoRA-adapted AASIST2 setting, architectural modifications that are beneficial in isolation can lead to performance degradation when combined.
- 3) We show that LearnT achieves the best mean EER, while concat-based scoring combined with multi-temperature routing yields the lowest cross-seed standard deviation.

II. RELATED WORK

a) Graph Attention Architectures for Audio Anti-Spoofing.: Building on graph attention networks [17] and early GAT-based anti-spoofing systems [18], AASIST [19] introduced heterogeneous graph attention networks to audio anti-spoofing, jointly modeling spectral and temporal graph branches with product-based symmetric scoring and a fixed softmax temperature. AASIST2 [7] improved the AASIST architecture with Res2Net-style residual blocks and additive-margin softmax, but the internal graph attention mechanism remained largely inherited from AASIST. Subsequent variants explored modifications to this design, but without isolating individual factors. RawGAT-ST [20] replaced the entire GAT module with a raw waveform GAT encoder and spectro-temporal graph attention, altering both input representation and attention mechanism simultaneously. Raw waveform front ends and augmentation have also been explored for anti-spoofing [21], [22]. The scalable AASIST line of work [12] substituted the GAT layers with multi-head attention, and

AASIST3 [13] evaluated HS-GAL branches with several fixed attention temperatures (e.g., $T = 100, 150$, and 200), treating temperature as a manually selected hyperparameter rather than a trainable parameter. SpAArSIST [23] recently simplified the HS-GAL stack-node attention, finding that temperature scaling did not improve their sparsified variant; however, this work examined attention simplification at the backend-pooling level rather than a controlled factorization of the original AASIST2 graph attention layer. Across these efforts, little prior work has decomposed that layer’s design choices into separately testable dimensions, and evaluated each under a common controlled protocol. This gap motivates our controlled decomposition of scoring symmetry, temperature learnability, and routing granularity as independent residual additions to a fixed baseline.

b) Scoring, Temperature, and Routing in Graph Attention.: Beyond the anti-spoofing domain, each of our three design axes has an established literature in graph representation learning. The element-wise-product scorer inherited by AASIST-style graph attention is necessarily symmetric ($e_{ij} = e_{ji}$), whereas GATv2 [14] showed that concat-based scoring with LeakyReLU broke such static ranking behavior and enabled query-adaptive attention weights, though this was demonstrated under full-training regimes on citation networks. Separately, temperature scaling is a standard calibration mechanism [24], and GATS [16] demonstrated that node-wise temperatures could be learned for GNN prediction calibration. Recent GAT variants have also explored learnable temperature as an explicit attention-sharpness control [25]. In the multi-expert direction, sparsely gated and switch-style expert routing [26], [27] motivated mixture-of-experts designs for graph neural networks [15], and multi-temperature attention mechanisms were explored for sequence models, but per-node temperature routing within a graph attention layer remained largely untested in any application domain. More broadly, recent work established that learnable scalar parameters could improve performance through their effect on optimization dynamics rather than their final values: Agarwala et al. [28] showed that the softmax temperature governed the transition from lazy to feature-learning regimes during training, and Wang et al. [29] demonstrated that scale vectors in LLMs were expressively redundant yet indispensable, improving optimization through a self-amplifying preconditioning effect. While each of these mechanisms was individually motivated and validated in its original context, none has been systematically evaluated in the specific setting where they interact: frozen pretrained SSL weights with trainable LoRA adapters and limited trainable capacity.

c) LoRA Adaptation of Self-Supervised Speech Front-Ends.: Self-supervised speech models such as Wav2Vec 2.0 [3], [30], [31], XLS-R [4], [6], WavLM [5], [32], and HuBERT [33] have become strong front-ends for spoofing and audio deepfake detection. Parameter-efficient fine-tuning provides a general framework for adapting large pretrained models with a small trainable parameter budget [8], [34], and recent audio deepfake systems have adopted LoRA-

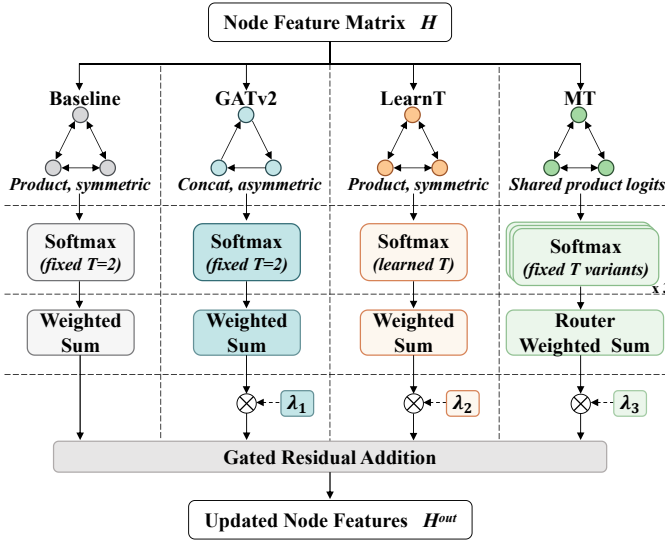


Fig. 2. Residual graph-attention framework for one AASIST2 graph-attention layer. The baseline uses symmetric element-wise-product scoring with fixed temperature $T_0 = 2.0$, while three gated residual branches instantiate the design axes: GATv2 for concat-based asymmetric scoring, LearnT for learnable softmax temperature, and MT for node-wise routing over fixed-temperature experts. Branch outputs are added to the preserved baseline path through learnable scalar gates (Eq. 7). In the scoring diagrams, bidirectional arrows denote symmetric scoring ($e_{ij} = e_{ji}$), whereas directed arrows denote asymmetric scoring ($e_{ij} \neq e_{ji}$), rather than changes to graph connectivity. The figure uses matrix-level notation for the layer input and output, while the equations use the corresponding node-level notation.

style adapters for cross-dataset generalization. Meta-learned LoRA [9] framed generalization across attacks as a domain-shift problem and applied MLDG-style meta-optimization to the LoRA weights. MoLEx [10] and AMULET [11] similarly explored adapter placement, expert fusion, and training strategies for cross-dataset generalization. Among these works, the downstream classifier, often a graph attention network, was treated as a fixed inheritance: LoRA rank, placement, and training objectives were systematically explored, while the internal scoring, temperature, and routing of the graph attention layers were carried over from the baseline without question. We address this gap by holding the LoRA configuration constant and systematically varying the graph attention internals to identify which design choices are consequential under the resulting capacity budget.

III. METHODOLOGY

A. Controlled LoRA-AASIST2 Setting

We study graph-attention design choices in AASIST2 under a controlled LoRA-adapted anti-spoofing pipeline. The base system uses an XLS-R-300M SSL encoder followed by the AASIST2 backend. Our intervention is restricted to the spectral and temporal graph-attention layers of AASIST2 (`gat_s` and `gat_t` in the implementation). Across all variants, we keep the SSL backbone, LoRA recipe, pooling module, classifier, and training protocol unchanged. This controlled setting reduces confounding from the front-end, adaptation

strategy, and training configuration, making the graph-attention interventions directly comparable. Figure 2 summarizes the controlled intervention scope: the XLS-R encoder and non-GAT backend components are held fixed, while the AASIST2 graph-attention layer is augmented with gated residual branches for the three design axes detailed in Section III-C and combined in Section III-D.

B. Baseline AASIST2 Graph Attention

Each targeted AASIST2 graph-attention layer, either spectral or temporal, maps a node feature matrix $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_N]^\top$, $\mathbf{h}_i \in \mathbb{R}^d$, to updated node representations. The baseline attention score between nodes i and j is computed with an element-wise-product scorer:

$$e_{ij}^{\text{base}} = \mathbf{a}^\top \tanh(\mathbf{W}_p(\mathbf{h}_i \odot \mathbf{h}_j)), \quad (1)$$

where $\mathbf{W}_p$ and $\mathbf{a}$ are learned parameters and $\odot$ denotes element-wise multiplication. A fixed temperature $T_0 = 2.0$ normalizes the scores before neighbor aggregation:

$$\alpha_{ij}^{\text{base}} = \text{softmax}_j \left(\frac{e_{ij}^{\text{base}}}{T_0} \right), \quad \tilde{\mathbf{h}}_i^{\text{base}} = \sum_j \alpha_{ij}^{\text{base}} \mathbf{h}_j. \quad (2)$$

Eq. (2) abstracts away the output projection, self-projection, batch normalization, and SELU activation used in the implementation; we denote the full baseline layer output by $\mathbf{h}_i^{\text{base}}$. This formulation reveals two inherited design properties. First, the score is symmetric because $\mathbf{h}_i \odot \mathbf{h}_j = \mathbf{h}_j \odot \mathbf{h}_i$, so $e_{ij}^{\text{base}} = e_{ji}^{\text{base}}$ before softmax normalization. Second, the attention temperature is fixed: its value is hard-coded to $T_0 = 2.0$ and shared globally across all nodes. We use these properties to derive three graph-attention design axes.

C. Graph-Attention Design Axes

We decompose the AASIST2 graph-attention design space into three independently testable axes. The first axis tests scoring symmetry, the second tests temperature learnability, and the third tests routing granularity. Each axis is instantiated as a residual branch and later combined through the gated framework in Section III-D.

Axis 1: Scoring Symmetry. The baseline element-wise-product scoring (Eq. 1) is symmetric, meaning $e_{ij} = e_{ji}$. To test whether asymmetric, query-adaptive scoring is beneficial under LoRA, we add a concat-based scoring branch [14]:

$$e_{ij}^{\text{gatv2}} = \mathbf{a}_g^\top \text{LeakyReLU}(\mathbf{W}_g(\mathbf{h}_i \parallel \mathbf{h}_j)), \quad (3)$$

In Eq. (3), $\parallel$ denotes concatenation and $\mathbf{W}_g, \mathbf{a}_g$ are learned branch-specific parameters. Since $(\mathbf{h}_i \parallel \mathbf{h}_j)$ generally differs from $(\mathbf{h}_j \parallel \mathbf{h}_i)$, this GATv2-style branch can produce $e_{ij}^{\text{gatv2}} \neq e_{ji}^{\text{gatv2}}$, breaking the symmetric scoring constraint of the baseline. The branch uses the same fixed temperature T_0 as the baseline and produces a full layer output $\mathbf{h}_i^{\text{gatv2}}$.

Axis 2: Temperature Learnability. The baseline fixes $T_0 = 2.0$ for both targeted graph-attention layers. Inspired by prior use of temperature scaling in graph attention [13], we instead make the temperature trainable by adding a branch that

replaces the constant with a learned scalar for each targeted layer:

$$T_\ell = \exp(\tau_\ell), \quad \tau_\ell \leftarrow \ln T_0 \text{ at initialization.} \quad (4)$$

The exponential parameterization in Eq. (4) ensures a positive temperature. This branch uses the same element-wise-product scoring form as the baseline, with independently initialized scoring and projection parameters, but replaces the fixed temperature with T_ℓ . At initialization, the branch uses the same temperature as the baseline; during training, T_ℓ can move away from T_0 when a different temperature is beneficial. The resulting full layer output is denoted $\mathbf{h}_i^{\text{lt}}$.

Axis 3: Routing Granularity. The baseline shares a single global temperature across all nodes. To test whether different nodes benefit from different fixed temperatures, we add a multi-expert branch with per-node routing [15]:

$$\alpha_{ij}^{(k)} = \text{softmax}_j \left(\frac{e_{ij}^{\text{mt}}}{T_k} \right), \quad T_k \in \{0.5, 2.0, 8.0\}, \quad (5)$$

In Eq. (5), all experts share the same element-wise-product scoring logits e_{ij}^{mt} but normalize them with different fixed temperatures. The k -th expert output is $\tilde{\mathbf{h}}_i^{(k)} = \sum_j \alpha_{ij}^{(k)} \mathbf{h}_j$. A node-wise router then mixes the expert outputs:

$$\mathbf{g}_i = \text{softmax} \left(\frac{\text{MLP}(\mathbf{h}_i)}{\rho} \right), \quad \tilde{\mathbf{h}}_i^{\text{mt}} = \sum_{k=1}^K g_{ik} \tilde{\mathbf{h}}_i^{(k)}, \quad (6)$$

In Eq. (6), $\mathbf{g}_i \in \mathbb{R}^K$ and ρ is a learnable router temperature initialized to 1.0. This routing is applied to the aggregated expert representations rather than to the raw attention scores. The resulting full layer output is denoted $\mathbf{h}_i^{\text{mt}}$.

D. Gated Residual Branch Framework

We test the three axes through a common residual interface rather than by directly replacing the baseline layer. This preserves the original AASIST2 attention path in every variant and provides an architecture-level degeneration path when a residual gate approaches zero. The output of each graph-attention layer is

$$\mathbf{h}_i^{\text{out}} = \mathbf{h}_i^{\text{base}} + \lambda_1 \mathbf{h}_i^{\text{gatv2}} + \lambda_2 \mathbf{h}_i^{\text{lt}} + \lambda_3 \mathbf{h}_i^{\text{mt}}, \quad (7)$$

where λ_1 , λ_2 , and λ_3 are learnable scalar gates for the GATv2, LearnT, and MT branches, respectively, and are initialized to a small positive value. A disabled branch is omitted from Eq. (7). This framework supports controlled single-axis comparisons and branch composition tests.

We define six branch configurations including the original baseline (Table I). The GATv2 name denotes the GATv2-style concat-scoring branch used to test the scoring-symmetry axis.

In addition to these configurations, we evaluate a FixedT-Res control with the same residual element-wise-product branch structure as LearnT but with temperature fixed at $T_0 = 2.0$. This control separates the effect of adding residual branch capacity from making the temperature trainable. We omit the MT+LearnT and GATv2+MT+LearnT combinations because MT and LearnT both manipulate temperature but through

TABLE I
BRANCH CONFIGURATIONS USED IN THE CONTROLLED STUDY. THE BASELINE USES THE ORIGINAL AASIST2 GRAPH-ATTENTION LAYER WITHOUT RESIDUAL BRANCHES. EACH NON-BASELINE CONFIGURATION ISOLATES ONE DESIGN AXIS OR TESTS A TWO-BRANCH COMBINATION UNDER THE GATED RESIDUAL FRAMEWORK (EQ. 7). ✓ = BRANCH ENABLED; – = DISABLED.

Variant	GATv2 (λ_1)	LearnT (λ_2)	MT (λ_3)
Baseline	–	–	–
GATv2	✓	–	–
LearnT	–	✓	–
MT	–	–	✓
GATv2+LearnT	✓	✓	–
GATv2+MT	✓	–	✓

TABLE II
SUMMARY OF DATASETS USED IN THE EXPERIMENTS. THE HORIZONTAL LINE SEPARATES TRAINING AND VALIDATION FROM EVALUATION.

Dataset	Usage	#Bonafide	#Spoofed
ASVspoof 2019 LA Train [35]	Training	2,580	22,800
ASVspoof 2019 LA Dev [35]	Validation	2,548	22,296
ASVspoof 2019 LA Eval [35]	In-domain	7,355	63,882
ASVspoof 2021 LA [36]	Out-of-domain	18,452	163,114
ASVspoof 2021 DF [36]	Out-of-domain	22,617	589,212
WaveFake [37]	Out-of-domain	13,100	91,700
In-the-Wild [38]	Out-of-domain	19,963	11,816

distinct mechanisms (fixed multi-temperature routing vs. a single learned scalar), and combining them would conflate two temperature effects within a single variant.

IV. EXPERIMENTS

A. Datasets

We evaluate on one in-domain and four out-of-domain test sets that differ from the training distribution along distinct axes. Table II summarizes the datasets used for training, validation, and evaluation.

All models are trained on the Logical Access (LA) partition of ASVspoof 2019 [35], [39], with the development set used for model selection. We evaluate on ASVspoof 2019 LA Eval [35] as the in-domain test set and on four out-of-domain sets: ASVspoof 2021 LA and DF [36], WaveFake [37], and In-the-Wild [38]. Together, these evaluation sets cover unseen attack types, codec and channel variation, neural-vocoder generation, and real-world public-media speech deepfakes.

B. Evaluation Metrics

We use a unified score-based protocol for all test sets. The primary metric is Equal Error Rate (EER), defined as the error rate at the operating point where the false acceptance and false rejection rates are equal. We call this metric Unified EER because the same score format and threshold-sweeping procedure are used for ASVspoof, WaveFake, and In-the-Wild, rather than using dataset-specific official scorers. For each seed, the Avg column is the unweighted mean of the five dataset-level Unified EER values; the reported mean and standard deviation are then computed over the five seeds.

As auxiliary metrics, we report Area Under the ROC Curve (AUROC) and Average Precision (AP).

C. Training Configuration

All LoRA-scale configurations share the same SSL front-end, LoRA setup, pooling module, classifier, optimizer, and training protocol; only the graph-attention intervention differs. The original XLS-R-300M pretrained backbone weights are kept frozen, while LoRA adapters and LayerNorm parameters are trainable. The LoRA adapters [8] with rank $r = 16$, scaling factor $\alpha = 16$, and dropout 0.05 are inserted into the query and value projections (q_{proj} , v_{proj}) of all 24 transformer layers. Across variants, the number of trainable parameters ranges from 2.51M to 2.57M, corresponding to approximately 0.79–0.81% of the full model. The learnable branch gates λ_1 , λ_2 , λ_3 are initialized to 0.05.

All reported models are trained with cross-entropy over additive-margin softmax (AM-Softmax) logits. Given feature vector $\mathbf{f}$ and class weight $\mathbf{w}_c$, the classifier uses normalized cosine logits $z_c = s \cos(\mathbf{f}, \mathbf{w}_c)$ for non-target classes and $z_y = s(\cos(\mathbf{f}, \mathbf{w}_y) - m)$ for the target class, with scale $s = 15$ and margin $m = 0.3$. Label smoothing, center loss, and feature-level mixup are disabled in the reported XLS-R experiments.

We use AdamW [40] with learning rate 10^{-4} and weight decay 10^{-4} . A warmup-cosine schedule is applied with 5% warmup and a minimum learning rate of 5% of the peak value. All models are trained for 30 epochs with batch size 16, single-precision (FP32) training, and 64,600-sample inputs (approximately 4 seconds at 16 kHz). Validation is performed every 2 epochs on ASVspoof 2019 LA Dev, and the checkpoint used for test evaluation is selected by the lowest development-set EER. Every configuration is trained independently under five random seeds {1234, 2345, 3456, 5678, 6789}. The seed controls parameter initialization, data shuffling, dataloader workers, and the main random number generators, enabling a direct assessment of training stability.

D. Main Results

Table III reports Unified EER across the five evaluation sets, and Figure 3 visualizes AUROC and AP averaged over the same sets. Lower EER is better, whereas higher AUROC and AP are better. In addition to the controlled variants from Table I, we include the FixedT-Res control to separate temperature learnability from residual branch capacity.

We additionally report official ASVspoof scorer results in Table IV. This evaluation uses the official ASVspoof scoring protocol and is restricted to the three ASVspoof test sets, since WaveFake and In-the-Wild do not provide ASVspoof official scoring files. Therefore, Unified EER over all five test sets remains our primary metric, while official ASVspoof EER serves as a protocol-specific consistency check and is not included in the five-set Avg column.

LearnT achieves the best average Unified EER among the LoRA-scale configurations, reducing the Baseline from

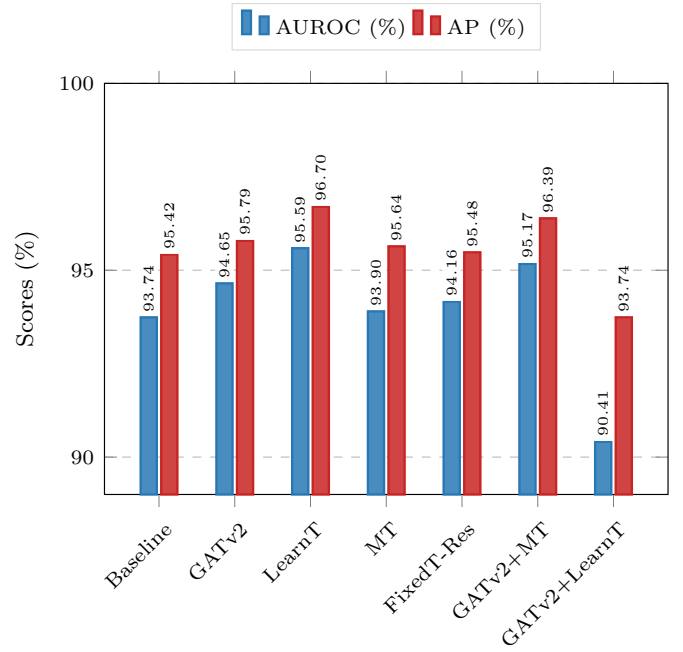


Fig. 3. Average AUROC and AP across the five evaluation sets. LearnT achieves the highest AUROC and AP, followed by GATv2+MT, consistent with the Unified EER ranking in Table III. Higher is better.

10.72% to 8.99%. GATv2+MT obtains the second-best average EER (9.23%) and the smallest seed-level standard deviation in the Avg column (0.19%). By contrast, MT alone is nearly unchanged from the Baseline (10.71% vs. 10.72%), and GATv2+LearnT degrades to 13.46%. The auxiliary metrics in Figure 3 follow the same ranking: LearnT achieves the highest average AUROC (95.59%) and AP (96.70%), followed by GATv2+MT (95.17% AUROC and 96.39% AP). The official ASVspoof results are consistent with this ranking: LearnT gives the best official average EER, and GATv2+MT remains the second-best LoRA-scale configuration.

E. Analysis of Model Variants

Unless noted otherwise, all relative improvements discussed in this section are computed based on the average Unified EER column of Table III.

a) *LearnT Gives the Strongest Single-Axis Gain.*: LearnT reduces the average Unified EER from 10.72% to 8.99%, corresponding to a 16.1% relative reduction over the Baseline. This is the largest mean improvement among the single-axis variants: GATv2 reduces average EER to 9.62% (10.2% relative reduction), whereas MT remains nearly unchanged at 10.71%. The FixedT-Res control reaches $10.23\% \pm 1.97$, which improves only modestly over the Baseline and remains 1.24 percentage points worse than LearnT. This comparison indicates that the LearnT gain is not explained by residual branch capacity alone; the learnable temperature contributes beyond adding another fixed-temperature branch.

b) *Branch Composition Is Non-Additive.*: The two tested branch combinations behave in opposite ways. GATv2+MT

TABLE III

UNIFIED EER (%) ACROSS FIVE TEST SETS. MEAN AND STANDARD DEVIATION ARE CALCULATED OVER FIVE RANDOM SEEDS. BEST AND SECOND-BEST RESULTS AMONG LoRA-SCALE CONFIGURATIONS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY. LOWER IS BETTER.

Method	ASV19 LA	ASV21 LA	DF21	WaveFake	In-the-Wild	Avg
Baseline	0.63±0.28	7.22±0.65	8.02±0.76	21.24±3.88	16.49±1.77	10.72±1.24
GATv2	0.61±0.21	7.22±0.88	7.48±1.14	17.77±2.30	15.04±2.33	9.62±1.18
LearnT	0.63±0.11	<u>6.81</u> ±0.70	6.58 ±1.03	<u>17.06</u> ±2.22	13.90 ±3.01	8.99 ±1.11
MT	0.65±0.26	7.75±1.50	9.19±4.66	19.24±5.49	16.73±6.27	10.71±3.42
FixedT-Res	0.55 ±0.11	7.54±1.47	8.29±2.70	19.61±1.89	15.18±4.21	10.23±1.97
GATv2+MT	<u>0.60</u> ±0.20	6.77 ±0.11	<u>7.05</u> ±0.27	16.47 ±0.82	15.28±0.78	<u>9.23</u> ±0.19
GATv2+LearnT	1.36±1.20	7.83±1.74	9.90±7.24	29.37±12.75	18.86±6.30	13.46±5.19

TABLE IV

OFFICIAL ASVSPOOF EER (%) COMPUTED WITH THE OFFICIAL ASVSPOOF SCORING PROTOCOL AND AVERAGED OVER ASVSPOOF 2019 LA, ASVSPOOF 2021 LA, AND ASVSPOOF 2021 DF. MEAN AND STANDARD DEVIATION ARE COMPUTED OVER FIVE RANDOM SEEDS. RELATIVE REDUCTION IS COMPUTED AGAINST THE BASELINE; NEGATIVE VALUES INDICATE DEGRADATION. LOWER EER IS BETTER.

Method	Official Avg	Rel. Reduction
Baseline	2.88±0.36	–
GATv2	2.62±0.27	+9.0%
LearnT	2.38±0.22	+17.3%
MT	3.35±1.81	–16.3%
FixedT-Res	2.72±0.58	+5.4%
GATv2+MT	2.46±0.20	+14.4%
GATv2+LearnT	4.49±3.58	–56.0%

improves average EER to 9.23% and reduces the Avg standard deviation to 0.19%, even though MT alone provides no average gain and has the largest single-axis standard deviation (3.42%). In contrast, GATv2+LearnT combines two individually beneficial branches but increases average EER to 13.46%, a 25.6% relative degradation from the Baseline. The degradation appears on every evaluation set: ASV19 LA increases from 0.63% to 1.36%, ASV21 LA from 7.22% to 7.83%, DF21 from 8.02% to 9.90%, WaveFake from 21.24% to 29.37%, and In-the-Wild from 16.49% to 18.86%. These results show that graph-attention branches do not combine additively under the fixed LoRA setting.

c) Improvements Concentrate on Out-of-Domain Sets.:

LearnT leaves the in-domain ASVspoof 2019 LA EER essentially unchanged (0.63% for both Baseline and LearnT), but improves all four out-of-domain sets. The out-of-domain average EER decreases from 13.24% for the Baseline to 11.09% for LearnT, a 16.3% relative reduction. The strongest absolute reduction occurs on WaveFake, where EER drops from 21.24% to 17.06%. GATv2+MT is less accurate on average than LearnT, but it is the most stable configuration: its Avg standard deviation is 0.19% compared with 1.24% for the Baseline and 1.11% for LearnT, and its out-of-domain coefficient of variation is 3.90% compared with 11.88% for the Baseline. We therefore treat GATv2+MT as a stability-oriented composition rather than the best mean-EER variant.

V. DISCUSSION AND IMPLICATIONS

The results show that the AASIST2 backend remains an important design lever in LoRA-adapted SSL anti-spoofing systems. Recent systems draw on SSL encoders, graph-attention backends, and parameter-efficient adaptation [2], [6]–[11], which makes it difficult to attribute gains to a specific component. By fixing the encoder, adapter placement, classifier, optimizer, and data protocol, our experiments isolate graph-attention design under the same adaptation budget. The resulting spread is substantial: among LoRA-scale variants, the best and worst average Unified EER differ by 4.47 percentage points despite nearly identical trainable parameter counts. Thus, the AASIST2 graph-attention layer is not merely an implementation detail behind the SSL front-end, but a meaningful source of cross-domain behavior.

Among the tested axes, the strongest positive result comes from temperature learnability rather than added branch capacity. LearnT outperforms both the Baseline and the FixedT-Res control, separating the effect of a trainable temperature from the effect of adding another residual graph-attention path. This aligns with work on calibration and optimization dynamics [16], [24], [28] and extends fixed-temperature anti-spoofing studies toward a trainable setting [13], [23]. The learned temperature remains close to the original AASIST2 value, averaging 1.96 after initialization at $T_0 = 2.0$, which suggests limited adaptive control over softmax sharpness rather than a different attention regime.

The branch-composition results provide the complementary caution. GATv2 and LearnT each improve average EER alone, but GATv2+LearnT degrades below the Baseline on every evaluation set; conversely, MT is weak as a single branch, yet GATv2+MT gives the second-best average EER and the lowest seed-level standard deviation. Backend attention should therefore be treated as a coupled system of scoring, normalization, and routing decisions. Since the AASIST2 graph topology is held fixed, these conclusions concern attention mechanisms rather than graph construction. Practically, LearnT is preferable when mean out-of-domain EER is the target, whereas GATv2+MT is preferable when seed-level stability is prioritized. Evaluations should report both out-of-domain performance and seed-level stability, in line with cross-dataset evaluation practice in audio deepfake detection [36], [38], [41]–[46].

VI. CONCLUSION

We presented a controlled study of graph-attention design choices in a LoRA-adapted XLS-R-300M + AASIST2 anti-spoofing system. By fixing the SSL front-end, LoRA configuration, classifier, and training protocol, we isolated three backend axes: scoring symmetry, temperature learnability, and routing granularity. The results show that graph-attention design remains important under a strict LoRA-scale parameter budget: LearnT achieves the best average Unified EER and reduces out-of-domain EER from 13.24% to 11.09%, whereas GATv2+MT provides the lowest seed-level standard deviation. This out-of-domain gain indicates improved cross-domain generalization rather than only better in-domain fitting. However, the degradation of GATv2+LearnT shows that backend interventions should be evaluated as interacting components rather than isolated plug-ins. Future work will extend this controlled analysis to other SSL encoders, adapter settings, and graph-based backends.

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